# Week 4 | GMM and EM

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**Reference and Credits:**

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# 1. Gaussian Mixture Model

For more complex distributions, we have what is called a Gaussian Mixture Model. A GMM is a probability distribution. It is a mixture of $K$ Gaussians, each of which is called a component.

## 1.1. Key Component

- ❖ Components
  - ➢ Each Gaussian distribution in the mixture is referred to as a "component."
  - ➢ Each component has its own mean $\mu$ and variance $\sigma^2$.
- ❖ Mixture Weights
  - ➢ Each Gaussian component has a weight $\pi_k$ indicating its contribution to the overall model, where $k$ indexes the components.
  - ➢ The weights satisfies the condition: $\sum_{i=1}^{K} \pi_k = 1$.
- ❖ Probability Density Function (PDF):

$$f(x;\theta) = \sum_{k=1}^{K} \pi_k \mathcal{N}\left(x; \mu_k, \sigma_k^2\right)$$

where
- $f$ is the PDF of the GMM.
- $\mathcal{N}$ is the PDF of a Gaussian distribution with mean $\mu_k$ and variance $\sigma_k^2$.
- $\pi_k$ is the "prior" contribution of the $k^{\text{th}}$ component and are called the mixture probabilities.

## 1.2. Sampling from GMM

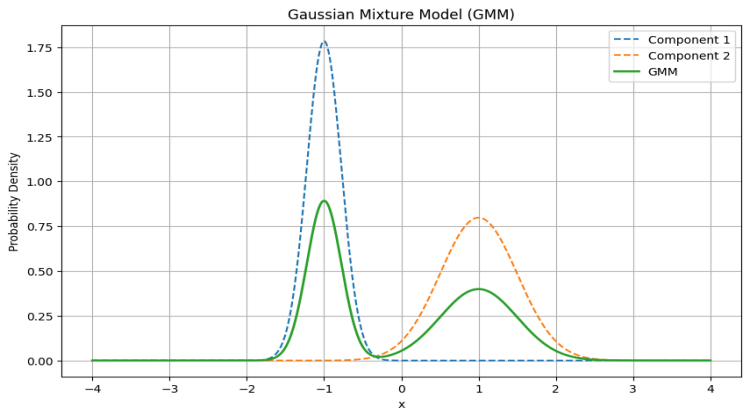
Sampling from a GMM is a two step process

- First choose a component $k$ with **prior** probability $\pi_k$.
- Sample a point from this Gaussian; the conditional density associated with this is $\mathcal{N}\left(x; \mu_k, \sigma_k^2\right)$.

## 1.3. Understanding of terms and Examples

**Example 1:**

$$\pi_1 = 0.5, \ \pi_2 = 0.5$$
$$\mu_1 = -1, \ \sigma_1^2 = \frac{1}{20}$$
$$\mu_2 = 1, \ \sigma_2^2 = \frac{1}{4}$$




$$f_X(x) = \pi_1 \cdot \mathcal{N}\left(x; \mu_1, \sigma_1^2\right) + \pi_2 \cdot \mathcal{N}\left(x; \mu_2, \sigma_2^2\right)$$

$$f_X(x) = 0.5 \times \frac{1}{\sqrt{\pi / 10}} \exp\left[-10(x+1)^2\right] + 0.5 \times \frac{1}{\sqrt{\pi / 2}} \exp\left[-2(x-1)^2\right]$$

**Understanding of terms**

Number of components
$k = 1, 2, \ \cdots K$

### 1.3.1. <u>P.M.F. of $Z$ (Prior Probability)</u>

$Z$ is the component indicator

$$f_Z(k) = P(Z = k) = \pi_k$$

There is $\pi_k$ chance that a data-point is going to be picked up from the $k^{th}$ component.

### 1.3.2. <u>Conditional Density</u>

$$f_{X \mid Z}(x \ \mid k) = \mathcal{N}\left(x; \ \mu_k, \sigma_k^2\right)$$

Given that you have picked a component , $f_{X \mid Z}(x \ \mid k)$ is the probability of observing the data-point $x$. How likely is it for $x$ to come from component $k$ ?

### 1.3.3. <u>Joint Density</u>

$$f_{X,Z}(x,k) = f_Z(k)\ .\ f_{X\mid Z}(x \mid k)$$

$$= \pi_k \cdot \mathcal{N}\Big(x;\ \mu_k, \sigma_k^2\Big)$$

Pick a component $k$ and sample a point $x$ from it.

### 1.3.4. <u>Marginal Density</u>

Density of G.M.M.

$$f_X(x) = \sum_{k=1}^{K} f_Z(k)\ .\ f_{X\mid Z}(x \mid k)$$

$$= \sum_{k=1}^{K} \pi_k \cdot \mathcal{N}\Big(x;\ \mu_k, \sigma_k^2\Big)$$

$$= \sum_{k=1}^{K} f_{X,Z}(x,k)$$

The point could have come from any component. You have to sum over all $K$ components.

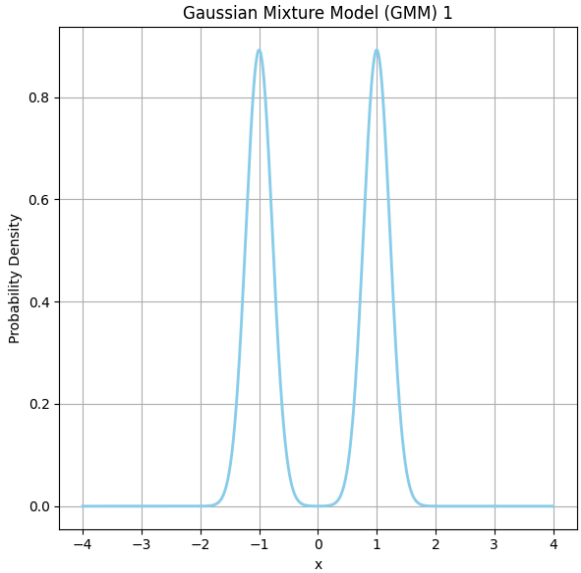
### 1.3.5. <u>Conditional Density</u>

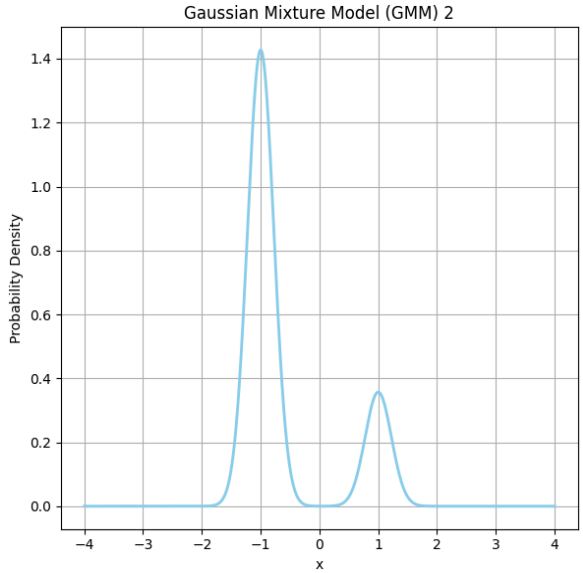
$$f_{Z\mid X}(k \mid x) = \frac{f_Z(k)\ .\ f_{X\mid Z}(x \mid k)}{\sum_{k=1}^{K} f_Z(k)\ .\ f_{X\mid Z}(x \mid k)}$$

### 1.3.6. <u>Parameters of the G.M.M.</u>

$\mu_1,\ \dots \mu_k$
$\sigma_1,\ \dots \sigma_k$ $\qquad \theta = [\pi, \mu, \sigma] \qquad 3K-1$ Parameters
$\pi_1 \dots \pi_k$

**Example 2:**



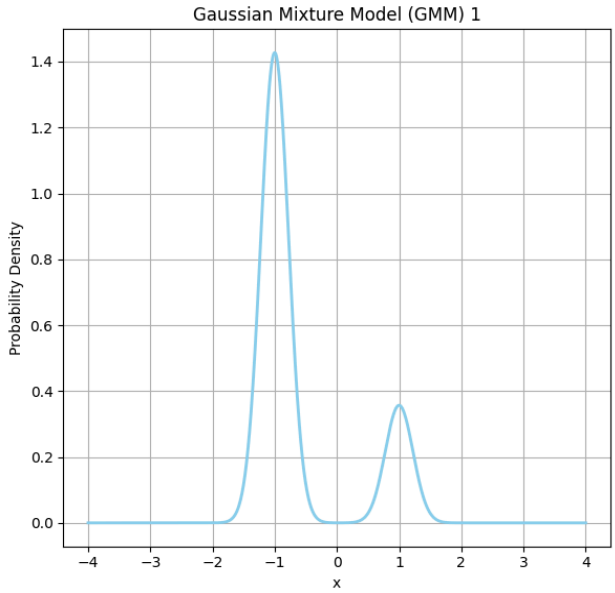



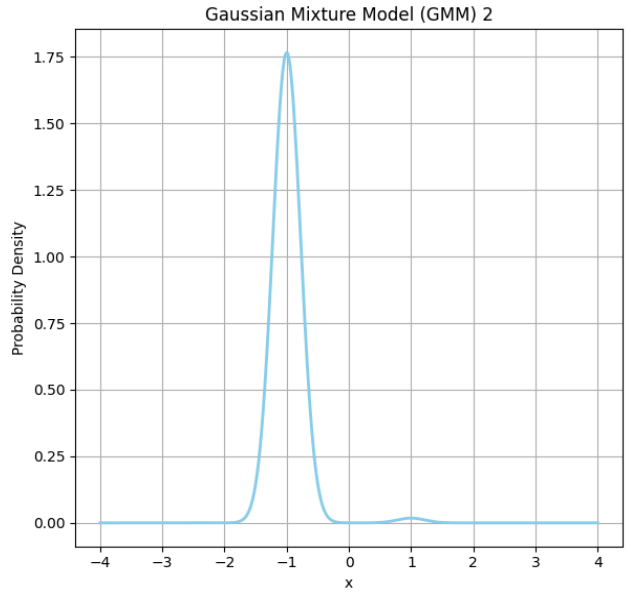

$$\pi_1 = 0.5,\ \pi_2 = 0.5$$
$$\mu_1 = 1,\ \sigma_1 = \frac{1}{20}$$
$$\mu_2 = -1, \sigma_2 = \frac{1}{20}$$

$$\pi_1 = 0.8,\ \pi_2 = 0.2$$
$$\mu_1 = 1,\ \sigma_1 = \frac{1}{20}$$
$$\mu_2 = -1, \sigma_2 = \frac{1}{20}$$

Here $z = k$ is some sense of prior.

**Example 3:**

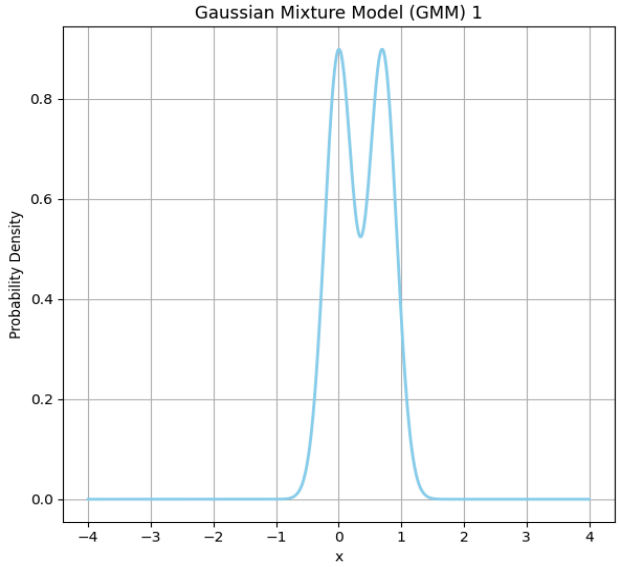


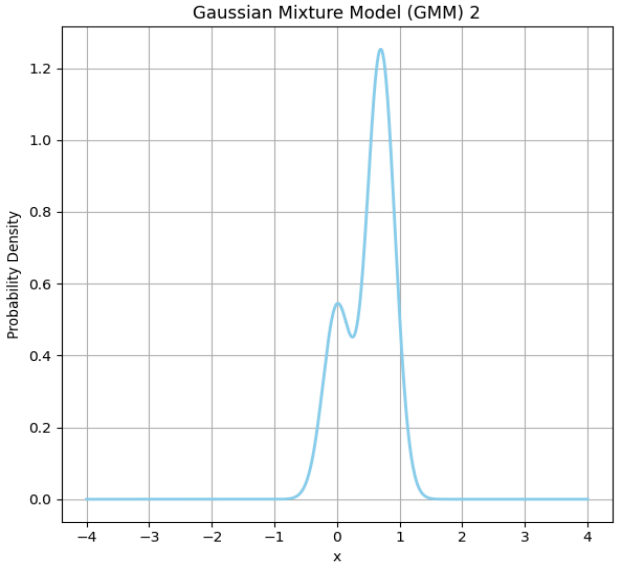




$$\pi_1 = 0.8,\ \pi_2 = 0.2$$
$$\mu_1 = 1,\ \sigma_1 = \frac{1}{20}$$
$$\mu_2 = -1, \sigma_2 = \frac{1}{20}$$

$$\pi_1 = 0.99,\ \pi_2 = 0.01$$
$$\mu_1 = 1,\ \sigma_1 = \frac{1}{20}$$
$$\mu_2 = -1, \sigma_2 = \frac{1}{20}$$

**Example 4:**




$$\pi_1 = 0.5,\ \pi_2 = 0.5$$
$$\mu_1 = 0,\ \sigma_1 = \frac{1}{20}$$
$$\mu_2 = 0.7, \sigma_2 = \frac{1}{20}$$

$$\pi_1 = 0.3,\ \pi_2 = 0.7$$
$$\mu_1 = 0,\ \sigma_1 = \frac{1}{20}$$
$$\mu_2 = 0.7, \sigma_2 = \frac{1}{20}$$

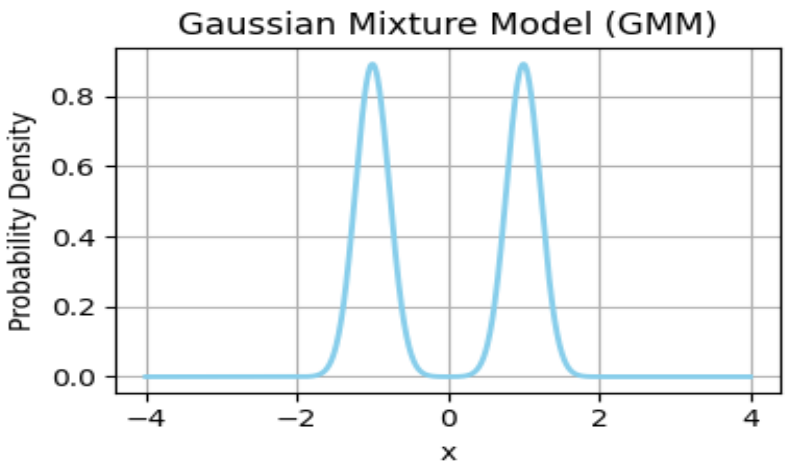
## 2. Prior and Posterior

$$f_Z(1) = \pi_1$$
$$f_Z(2) = \pi_2$$

$$\begin{aligned} f_{Z|X}(k \mid x) &= \frac{f_{X,Z}(z,k)}{f_X(x)} \\ &= \frac{f_Z(k) \cdot f_{X|Z}(x \mid k)}{\sum_{k=1}^{K} f_Z(k) \cdot f_{X|Z}(x \mid k)} \\ &= \frac{\pi_1 \cdot \mathcal{N}\left(x; \mu_1, \sigma_1^2\right)}{\pi_1 \cdot \mathcal{N}\left(x; \mu_1, \sigma_1^2\right) + \pi_2 \cdot \mathcal{N}\left(x; \mu_2, \sigma_2^2\right)} \end{aligned}$$

### 2.1. Example

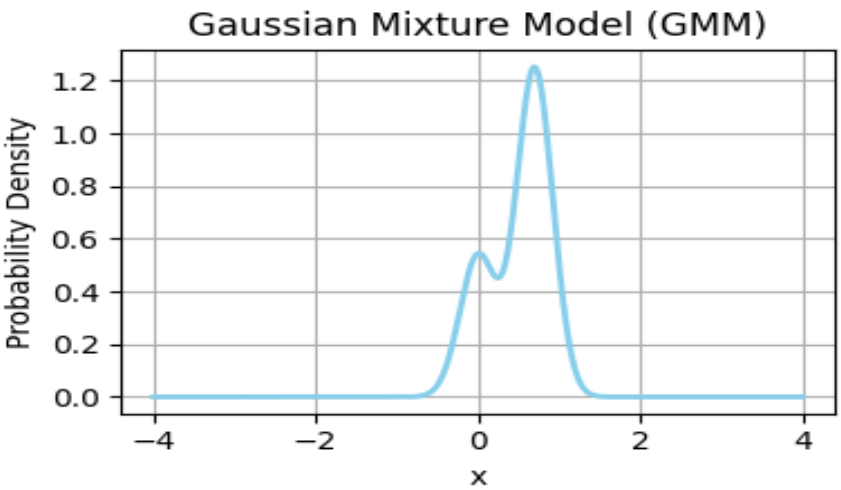
**Example 1:**




| $x$ | $f_{X\|Z}(x \mid k)$ |
|---|---|
| $-0.1$ | 0.98 |
| 0 | 0.5 |
| 0.1 | 0.02 |

$$\pi_1 = 0.5, \ \pi_2 = 0.5$$
$$\mu_1 = 1, \ \sigma_1 = \frac{1}{20}$$
$$\mu_2 = -1, \sigma_2 = \frac{1}{20}$$

**Example 2:**




$$\pi_1 = 0.3, \ \pi_2 = 0.7$$
$$\mu_1 = 0, \ \sigma_1 = \frac{1}{20}$$
$$\mu_2 = 0.7, \sigma_2 = \frac{1}{20}$$

| $x$ | $f_{X\|Z}(x \mid k)$ |
|---|---|
| 0.1 | 0.93 |
| 0.2 | 0.78 |
| 0.3 | 0.46 |
| 0.4 | 0.18 |
| 0.5 | 0.05 |

**The reason of confusion:** The farther the mean are, the better it is to make distinguish between two component. The closer the mean are, the greater potential for confusion. The overlapping would be reduced. But it does not only depend upon mean, the mixture probability also governs it.

## 3. Estimating the Parameters

Given the dataset

$$D = \{x_1, x_2, \cdots x_n\}$$

sampled from GMM with i.i.d. assumption and $K$ components.

Total parameters to estimate - $\theta = [\pi, \mu, \sigma] \quad 3K - 1$

Density function of GMM

$$f_X(x) = \sum_{k=1}^{K} \pi_k \cdot \mathcal{N}\left(x; \mu_k, \sigma_k^2\right)$$

**Likelihood function**

$$\begin{aligned} L(\theta; D) &= \prod_{i=1}^{n} f_X(x_i; \theta) \\ &= \prod_{i=1}^{n} \sum_{k=1}^{K} \pi_k \cdot \mathcal{N}\left(x; \mu_k, \sigma_k^2\right) \end{aligned}$$

### 3.1. Log Likelihood

$$\begin{aligned} l(\theta; D) &= \log \prod_{i=1}^{n} f_X(x_i; \theta) \\ &= \sum_{i=1}^{n} \log f_X(x_i; \theta) \\ &= \sum_{i=1}^{n} \log \left[ \sum_{k=1}^{K} \pi_k \cdot \mathcal{N}\left(x; \mu_k, \sigma_k^2\right) \right] \end{aligned}$$

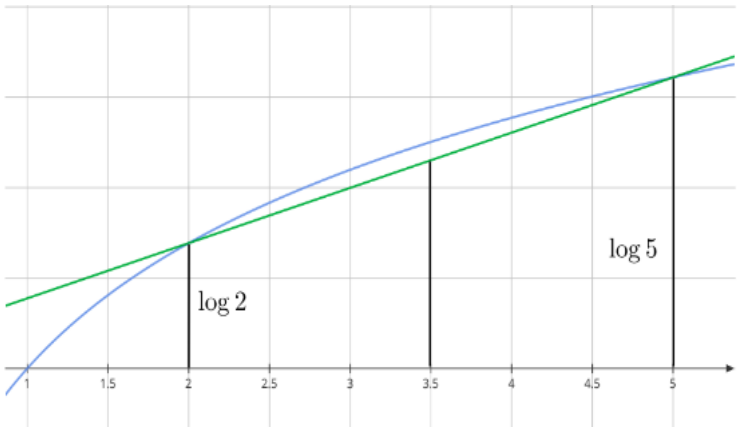
**Problem:** Summation inside $\log$.

## 4. Jensen's Inequality

$$\log(\lambda_1 x_1 + \lambda_2 x_2) \geqslant \lambda_1 \log(x_1) + \lambda_2 \log(x_2)$$

Condition: $\lambda_1 + \lambda_2 = 1$

$$\log\left(\frac{1}{2} \cdot 2 + \frac{1}{2} \cdot 5\right) \geqslant \frac{1}{2} \log 2 + \frac{1}{2} \log 5$$




### 4.1. $\lambda_k^i$

$$l(\theta; D) = \sum_{i=1}^{n} \log\left[\sum_{k=1}^{K} \pi_k \cdot \mathcal{N}\left(x; \mu_k, \sigma_k^2\right)\right]$$

$$= \sum_{i=1}^{n} \log\left[\sum_{k=1}^{K} \lambda_k^i \cdot \frac{\pi_k \cdot \mathcal{N}\left(x; \mu_k, \sigma_k^2\right)}{\lambda_k^i}\right]$$

$$\geqslant \sum_{i=1}^{n} \sum_{k=1}^{K} \lambda_k^i \log\left[\frac{\pi_k \cdot \mathcal{N}\left(x; \mu_k, \sigma_k^2\right)}{\lambda_k^i}\right]$$

$$= \sum_{i=1}^{n} E_{\lambda^i}\left[\log\left[\frac{\pi_k \cdot \mathcal{N}\left(x; \mu_k, \sigma_k^2\right)}{\lambda_k^i}\right]\right]$$

| $\lambda$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 0.5 | 0.1 | 0.3 | 0 |
| 2 | 0.5 | 0.7 | 0.5 | 1 |
| 3 | 0 | 0.2 | 0.2 | 0 |

$$\sum_{k=1}^{K} \lambda_k^i = 1$$

# 5. EM Algorithm

We can use MLE to estimate the parameters. But we don't have a closed form solution. Thankfully, we have an iterative approach to parameter estimation called the EM algorithm. This algorithm makes use of some intermediate variables that help in parameter estimation:

$$\lambda_k^i$$

Points to note:

- $i$ : corresponds to index of the data-point
- $k$ : corresponds to index of the component
- $\lambda_k^i$ can be interpreted as a conditional probability
  - $0 \leqslant \lambda_k^i \leqslant 1$
  - $\sum_{k=1}^{K} \lambda_k^i = 1$ for all $i$
- There are $Kn$ such variables

The parameters are collectively referred to as $\theta = [\pi, \mu, \sigma]$. We keep improving our estimate of $\theta$ in each step. There are two steps in the algorithm:

- E-step: update the values for $\lambda$ using the current values of $\theta$.
- M-step: update the values of $\theta$ using the newly found values of $\lambda$.

## 5.1. Procedure

<u>Initialization</u>

$$\theta^{(0)} = \begin{cases} \mu_1^{(0)}, \ \cdots, \mu_K^{(0)} \\ \left(\sigma_1^2\right)^{(0)}, \ \cdots, \left(\sigma_K^2\right)^{(0)} \\ \pi_1^{(0)}, \ \cdots, \pi_K^{(0)} \end{cases} \qquad \theta^{(t)} = \begin{cases} \mu_1^{(t)}, \ \cdots, \mu_K^{(t)} \\ \left(\sigma_1^2\right)^{(t)}, \ \cdots, \left(\sigma_K^2\right)^{(t)} \\ \pi_1^{(t)}, \ \cdots, \pi_K^{(t)} \end{cases}$$

<u>Convergence criterion</u>

When successive iterates become smaller than some $\epsilon$

$$\|\theta^{(t+1)} - \theta^{(t)}\| < \epsilon$$

**Until Convergence**

<u>E-step</u>

$$\lambda_k^i = f_{Z|X}(k \mid x)$$

$\lambda_k^i$ is the contribution of the $k^{th}$ component to the point $x_i$ given that we have observed the point $x_i$. It represents the posterior probability of $Z$ given $X$, that is $P(z_i = k \mid X = x_i)$. Using the Bayes' theorem:

$$\lambda_k^i = \frac{P(z_i = k) \cdot P(X = x_i \mid z_i = k)}{P(X = x_i)}$$

$$= \frac{\pi_k \cdot \mathcal{N}\left(x_i; \mu_k, \sigma_k^2\right)}{\sum_{k=1}^{K} \pi_k \cdot \mathcal{N}\left(x_i; \mu_k, \sigma_k^2\right)}$$

Here we use the current values of $\pi_k, \mu_k, \sigma_k^2$ to estimate $\lambda_k^i$.

<u>M-step</u>

Use the value of $\lambda_k^i$ obtained in E-step to update the values of $\pi_k, \mu_k, \sigma_k^2$.

$$\mu_k = \frac{\sum_{i=1}^{n} \lambda_k^i \cdot x_i}{\sum_{i=1}^{n} \lambda_k^i} \qquad \sigma_k^2 = \frac{\sum_{i=1}^{n} \lambda_k^i \cdot (x_i - \mu)^2}{\sum_{i=1}^{n} \lambda_k^i}$$

$$\pi_k = \frac{\sum_{i=1}^{n} \lambda_k^i}{n}$$

## 5.2. EM v/s K-means

| EM Algorithm | K-means |
|---|---|
| E-step: $\lambda_k^i = f_{Z\|X}(k \mid x)$ | Reassignment-step: $\lambda_k^i = \begin{cases} 1 & x_i \text{ is closest to } \mu_k \\ 0 & \text{otherwise} \end{cases}$ |
| M-step: $\mu_k = \dfrac{\sum_{i=1}^{n} \lambda_k^i \cdot x_i}{\sum_{i=1}^{n} \lambda_k^i}$ <br> $\sigma_k^2 = \dfrac{\sum_{i=1}^{n} \lambda_k^i \cdot (x_i - \mu)^2}{\sum_{i=1}^{n} \lambda_k^i}$ $\quad \pi_k = \dfrac{\sum_{i=1}^{n} \lambda_k^i}{n}$ | Compute mean: $\mu_k = \dfrac{\sum_{i=1}^{n} \lambda_k^i \cdot x_i}{\sum_{i=1}^{n} \lambda_k^i}$ |

## 5.3. Some final points about EM

- Under certain conditions Jensen's inequality becomes an equality. The **E-step** is all about exploiting this equality.
- After touching the main likelihood function, **M-step** tries to maximize our parameters $\mu, \sigma, \pi$ using the $\lambda$ values which we got at **E-step**.
- **EM** algorithm is soft clustering and **K-mean** is hard clustering.
- **EM** algorithm converges to **K-means**. (in context to Gaussian)